

Jeffrey Blay

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Professional Summary: Applied Science Ph.D. Candidate in Geospatial Data Science with 5+ years of experience in spatial AI, remote sensing, machine learning/computer vision, and geospatial data engineering. Builds scalable geospatial analytic solutions that transform geospatial datasets into actionable insights for evidence-based decision-making and address complex societal challenges.

Education

Ph.D. Applied Science and Technology, Geospatial Data Science (Expected December 2026)

North Carolina A&T State University

Relevant Coursework: Machine Learning and Data Mining, Big Data Analytics, Advanced Geospatial Analysis, Neural networks, Statistical Methods, Multivariate Statistics for Engineers

Master of Environmental Science, Environmental Data science for Urban sustainability (May 2023)

Yale University, School of the Environment

Relevant Coursework: Advanced Remote Sensing, Statistics for Environmental Sciences, Data Exploration & Analysis, Environmental Data Science, Introductory Machine Learning, Urban Landuse Clinic

Bachelor of Arts, Geography with Political Science (May 2019)

University Of Ghana • GPA: 3.81/4.0

Professional Experience

Geospatial Research Engineer

June 2026 – Present

Remote Sensing & GIS Lab, North Carolina A&T State University

- Building a QGIS plugin that deploys CNN and transformer-based flood-depth models for efficient on-demand inference by end users.
- Assessing disaster impacts on crop-specific land cover and vulnerable communities using federal datasets and Google Earth Engine.
- Leading a peer-reviewed journal publication on developing physics-informed AI framework for accurate urban flood modeling and infrastructure vulnerability assessment.

Geospatial Data Science Research Assistant, NASA Environmental Data Project

September 2024 – May 2026

GEMS Institute, North Carolina A&T State University

- Developed and optimized CNN- and transformer-based models (Swin-UNet, SegFormer, UNet, Attention UNet, UNet++) in PyTorch for urban flood-depth prediction, achieving ~ 0.19 ft average MAE.
- Developed a physics-informed deep learning framework incorporating domain-knowledge constraints to improve the physical consistency of urban flood-depth modeling by ~ 84% compared with conventional deep learning models.
- Built geospatial ETL pipelines with python (rasterio, GDAL, pandas) to ingest, align, and preprocess ~347 million pixel-level records from multi-source raster data for training deep learning models.
- Published *ezprocess*, a python library (PyPI/GitHub) that cut geospatial data-prep time by ~85% across ML workflows.
- Led a 4-person research team to design and publish a novel flood-depth benchmark dataset for reproducible flood-depth inundation modeling in urban environments.
- Managed code, experiments, and library version control using Git and GitHub to enable collaborative development across 2 research teams.

Graduate Research Assistant, NSF Multimodal Data Fusion Project

September 2023 - August 2024

Remote Sensing & GIS Lab, North Carolina A&T State University

- Built ETL workflows using Python (NumPy and pandas) to prepare ~700K geospatial point records for Machine Learning flood prediction models, reducing data preparation process time by 60%.
- Evaluated Random Forest and XGBoost flood prediction models using train-test splits, cross-validation, and performance metrics to assess predictive accuracy and model reliability.

- Collaborated with a 5-member team to develop and publish a multi-source geospatial dataset for inundated vegetation modeling; designed automated pipelines to preprocess and integrate multi-sensor datasets using ArcPy in ArcGIS Pro.
- Developed automated geospatial data quality assurance (QA/QC) and validation tests to verify spatial accuracy, attribute consistency, and data integrity prior to publication.
- Built Python web-scraping workflows using BeautifulSoup and SQLite to collect, curate, and organize ~1,000 web images for image classification tasks
- Developed a PyTorch framework to train and evaluate image classification models, including GoogLeNet and ResNet18, for automated flood classification of crowdsourced images.
- Conducted a technical assessment of GAN-based approaches for flood mapping using Earth observation datasets, analyzing their applicability to geospatial image generation and flood mapping.

GIS & Community Engagement Manager
Urban Resources Initiative

May – August 2023

- Managed 7 community greenspace groups, translating neighborhood priorities into weekly project plans and implementation strategies to advance community-led environmental initiatives.
- Applied ArcGIS Pro and QGIS to develop spatial analyses, maps, and visualizations that translated community needs into actionable greenspace planning concepts.
- Delivered GIS-based visualizations to support stakeholder decision-making, helping community groups assess site-specific opportunities and refine proposed greenspace improvements.

Teaching Fellow

January 2023 – May 2023

Real-world Environmental Data Science Course, Yale University

- Mentored 12 students in Python-based environmental data science, troubleshooting data processing, analysis, and programming challenges across assignments and projects.
- Supported a collaborative data science workflow using Google Colab, GitHub, and GitHub Classroom, helping students develop reproducible coding practices and manage project-based analyses.

Research Assistant, NASA Environmental Justice Project
Hixon Center For Urban Ecology, Yale University

September 2022 – May 2023

- Analyzed multispectral satellite imagery and geospatial data at the census-block level to identify urban communities vulnerable to extreme heat and urban heat island (UHI) effects.
- Developed ArcGIS Pro workflows using ArcPy and SQL queries to process and analyze block-level land surface temperature data, supporting spatial assessment of urban heat exposure.
- Conducted time-series statistical analysis of urban heat and tree-cover disparities using R (sp, dplyr, ggplot2), producing maps and visualizations to characterize environmental inequities across urban communities.

Research Fellow, TRI Research Fellowship
Tropical Research Institute (TRI), Yale University

May 2022 – May 2023

- Trained a U-Net regression model to predict building composition across metropolitan Ghana from landsat satellite imagery, achieving an R^2 of ~75% on held-out test data.
- Processed VIIRS nighttime light data in Google Earth Engine to analyze spatial patterns in electricity distribution and identify areas of limited infrastructure access.
- Applied spatial and geostatistical analysis to identify and map electricity infrastructure gaps across built-up areas, supporting assessment of underserved communities.
- Designed and deployed an Integrated Field Mapping and Real-Time Monitoring System using ArcGIS Pro, ArcGIS Online, Field Map, and a Dashboard to collect, monitor, and analyze 300+ ground-truth building observations, improving data collection accuracy by 90%.

Geospatial Research Assistant, Urban Africa Project
Seto Lab, Yale University

September 2021 - May 2022

- Developed and optimized Google Earth Engine workflows to acquire and preprocess millions of satellite pixels for large-scale urban analysis, improving the scalability of downstream geospatial modeling.
- Trained satellite image segmentation models in Python across 800+ image tiles on a high-performance Linux computing cluster, implementing scalable batch-processing pipelines for large-scale geospatial analysis.

- Developed custom R and Python workflows for geospatial data processing, statistical analysis, and visualization, supporting analysis of urban environmental and spatial patterns.

Geospatial Data Analyst

September 2020 - June 2021

National Census Secretariat, Ghana Statistical Service

- Performed geospatial data quality assurance and control for national census datasets, validating spatial consistency, boundary alignment, coordinate reference systems (CRS), geometry, and attribute integrity across point, line, and polygon datasets.
- Restructured and corrected Supervisory Area (SA) boundary datasets by identifying and resolving spatial errors, misaligned boundaries, redundant features, and incorrect attributes to improve the accuracy and consistency of census geospatial data.
- Cross-validated point, line, and polygon datasets to reconcile spatial discrepancies and ensure consistency between census geographic boundaries and associated field data.
- Conducted spatial analysis to identify difficult-to-enumerate areas during national census operations, supporting targeted allocation of field resources and producing near-real-time spatial coverage analytics with a 15-member team to monitor enumerator operations.

Teaching / Research Assistant

September 2019 - August 2020

Remote Sensing/GIS Lab, University of Ghana

- Delivered Remote Sensing and GIS tutorial sessions for 150 undergraduate students, providing hands-on instruction in geospatial concepts, spatial analysis, and GIS workflows.
- Mentored students through GIS laboratory projects and group assignments, providing technical guidance on geospatial data processing, analysis, and visualization.
- Designed, administered, and graded GIS assignments and group projects, assessing students' application of remote sensing and geospatial analysis techniques.

Physical Planning Intern

May - August 2018

Physical Planning Department, Awutu Senya East Municipal Authority

- Conducted field inspections with Planning Officers to assess land-use activities against zoning regulations, planning codes, and approved development plans.
- Applied GIS techniques and aerial imagery to analyze land-use patterns, support zoning assessments, and coordinate development activities across the municipality.
- Maintained and developed spatial and attribute data supporting land title registration and development permit reviews, ensuring consistency between property records and planning requirements.
- Supported development permit approvals by reviewing submitted plans and land-use information for compliance with applicable zoning and planning regulations.

Projects

GridVision MLOps - jeffreyblay.github.io/energy_demand_mlops

2026

Python · FastAPI · PostGIS · Supabase · GitHub Actions · MapLibre GL · Render

- End-to-end MLOps system forecasting US electricity demand with live data, retraining, model gating, production promotion, and 3D visualization.

PawPal+

2026

Python · Streamlit · GeminiAPI · RAG · Pytest · JSON

- Pet-care scheduling and RAG assistant that generates priority-based, conflict-aware daily plans, time-budget optimization, reminders, and personalized pet-care Q&A using Gemini.

Quickview Geodata Portal — jeffreyblay.github.io/quickview-geodata-portal

2026

Python · FastAPI · GeoPandas · Render Cloud · Leaflet.js · GitHub Pages

- Built an open-source geospatial platform for fast, interactive spatial data exploration from URL for different file formats.

Weather Alert Dashboard— jeffreyblay.github.io/climate-alert-dashboard

2026

- Built real-time weather monitoring platform integrating National Weather Service API; REST API with filtering, analytics, and risk-scoring across all 50 U.S. states.

Publications

Blay, J., Gebregziabher, Y., Jha, M. K., & Hashemi Beni, L. (2026). Inundation2Depth: A multi-source dataset for floodwater depth estimation in urban areas. *Data in Brief*, 64, 112347. <https://doi.org/10.1016/j.dib.2025.112347>

Blay, J., & Hashemi-Beni, L. (2025a). Geospatial and Deep Learning Approaches for Modeling Floodwater Depth in Urbanized Areas. *Remote Sensing*, 18(1), 60. <https://doi.org/10.3390/rs18010060>

Blay, J., & Hashemi-Beni, L. (2025b). Pixels to Insights: Deep Learning for Floodwater Depth Mapping in Settlement Areas. *IGARSS 2025 - 2025 IEEE International Geoscience and Remote Sensing Symposium*, 1140–1144. <https://doi.org/10.1109/IGARSS55030.2025.11242913>

Blay, J., & Hashemi-Beni, L. (2025c, April). *Advanced Geo-Data Analytics and AI for 3D Flood Mapping to Protect Built Assets*. ISPRS Geospatial Week 2025, Dubai,UAE.

Blay, J., Fawakherji, M., & Hashemi-Beni, L. (2024, September 5). *Flood impact risk mapping in settlement areas from a 3D perspective: A case study of hurricane matthew*. 2024 IEEE International Geoscience and Remote Sensing Symposium, Athens, Greece. <https://doi.org/10.1109/IGARSS53475.2024.10640634>.

Blay, J., Seto, K. C., & Chen, T.-H. K. (2024). *Dark Development: Satellite Analysis of Building Density and Electricity Provision in Ghana's Urban Areas*. SSRN. <https://doi.org/10.2139/ssrn.5038955>

Fawakherji, M., Anokye, M., **Blay, J.,** & Hashemi-Beni, L. (2025). Multi-Resolution Data Fusion for Resilient Flood Mapping. *IEEE Access*, 13, 202275–202294. <https://doi.org/10.1109/ACCESS.2025.3636732>

Fawakherji, M., **Blay, J.,** Anokye, M., Hashemi-Beni, L., & Dorton, J. (2025). DeepFlood for Inundated Vegetation High-Resolution Dataset for Accurate Flood Mapping and Segmentation. *Scientific Data*, 12(1), 271. <https://doi.org/10.1038/s41597-025-04554-3>

Owusu, A. B., Mensah, C. A., **Blay, J.,** & Fynn, I. E. M. (2023). Urban Growth and Land Surface Temperature Dynamics: Lessons from Ghana. *Theoretical and Empirical Research in Urban Management*, 18(3).

Syum Gebre, T., Endale Ashebir, S., **Blay, J.,** Anokye, M., Pandey, V., & Hashemi-Beni, L. (2025). Real-Time Traffic Insights With Physics-Informed Neural Networks: Integrating the Aw-Rascle Model and LLMs. *IEEE Access*, 13, 181246–181266. <https://doi.org/10.1109/ACCESS.2025.3618282>

Technical Skills

Programming : Python, R, SQL, JavaScript

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Supervised Learning (Logistic Regression, Random Forest, XGBoost, LightGBM, UNet, Attention-UNet, Swin-UNet,Segformer, Physics-Informed Neural Networks) Model Evaluation (Huber Loss, RMSE, MAE, SSIM, Precision, Recall, AUC-ROC, Physics-Informed Loss), Unsupervised Learning (K-means)

Geospatial & Remote Sensing: ArcGIS Pro, ArcGIS Online, ArcPy, QGIS, Google Earth Engine, GDAL, ENVI, SNAP, Spatial Analysis, Remote Sensing, Geospatial Data QA/QC, PostGIS, MapLibre GL

Data Engineering: ETL Pipelines, Data Preprocessing, Web Scraping, Apache Airflow, RDBMS

Statistics & Analytics: Spatial Statistics, Geostatistics, Time-Series Analysis, Statistical Modeling, Multivariate Analysis, Hypothesis Testing

Cloud, DevOps & Infrastructure: AWS (S3, EC2), Docker, CI/CD, Git, GitHub, GitHub Actions, HPC, Render

Web & APIs: FastAPI, RESTful APIs, Streamlit, Vercel, HTML/CSS

Databases: PostgreSQL, MySQL, IBM DB2

Visualization: Matplotlib, Seaborn, Power BI, Tableau, R Shiny, ArcGIS Dashboards, ArcGIS StoryMaps, Google Data Studio

Conferences & Presentations

- 2025 IGARSS (International Geoscience and Remote Sensing Symposium) Conference** **August 7, 2025.**
- **Presentation Topic:** Pixels to Insights: Deep learning for Floodwater Depth Mapping in Settlement Areas.
- 2025 ISPRS Geospatial Week** **April 10, 2025.**
- **Presentation Topic:** Advanced Geo-Data Analytics and AI for 3D Flood Mapping to Protect Built Assets
- 2025 Annual ASPRS and Geo Week Conference** **February 12, 2025.**
- **Presentation Topic:** 3D Floodwater Depth Analytics with Big Geo-Data and cGAN.
- 2024 Annual AGU (American Geophysical Union) Conference** **December 11, 2024.**
- **Presentation Topic:** Urban Flood Depth Analytics: Leveraging Generative Algorithms and Remote Sensing for 3D Flood Depth Simulation.
- 2024 NCAUG (North Carolina ArcGIS Users Group) Conference** **September 24, 2024.**
- **Presentation Topic:** Leveraging AI and big geo-data for 3D flood depth mapping to safeguard-built assets.
- 2024 IGARSS (International Geoscience and Remote Sensing Symposium) Conference** **July 08, 2024.**
- **Presentation Topic:** Flood Impact Risk Mapping in Settlement Areas from a 3D Perspective: A Case Study of Hurricane Matthew.
- 2024 Annual ASPRS and Geo Week Conference** **February 14, 2024.**
- **Presentation Topic:** A 3D perspective of analysing the effect of Hurricane Matthew in small communities: A case study of Grifton, NC.
- 2023 Annual AGU (American Geophysical Union) Conference** **December 14, 2023.**
- **Presentation Topic:** Multi-channel deep learning-based architecture for flood detection and mapping
- Tropical Resource Institute(TRI) Symposium** **April 21, 2023.**
- **Presentation Topic:** Local Scale analysis of Urban Intensity and Electricity infrastructures in Ghana with multi-sensor satellite images and machine learning techniques; A case study of Metropolitan cities.
- 39th Yale School of the Environment(YSE) Research Day** **April 14, 2023.**
- **Presentation Topic:** Local Scale analysis of Urban Intensity and Electricity infrastructures in Ghana with multi-sensor satellite images and machine learning techniques; A case study of Metropolitan cities
- Yale School of the Environment(YSE) Confluence Talk Series** **March 24, 2023.**
- **Presentation Topic:** Local Scale analysis of Urban Intensity and Electricity infrastructures in Ghana with multi-sensor satellite images and machine learning techniques; A case study of Metropolitan cities

Honors & Awards

- Graduate Student Awards** **2026**
- Outstanding Doctoral Graduate Research Assistant Award for the College of Science and Technology - NC A&T State University
- Lidar Leader Award** **2024**
- Best Poster presentation - 2024 Annual ASPRS & Geo Week Conference

Grants & Fellowships

- Graduate Research Assistant Fellowship** **2023 - 2026**
- North Carolina A&T State University
- Teaching Fellowship** **2023**
- Yale School of the Environment(YSE)

TRI Research Endowment Fellowship Tropical Resource Institute, Yale University	2022
YSE need based and Merit Scholarship (Master's Study) Yale School of the Environment (YSE)	2021 – 2023

Leadership and Co-Curricular Activity

Secretary & Social Media Manager <i>ASPRS Student Chapter, North Carolina A&T State University</i>	2024 – 2026
- Managed the chapter's LinkedIn presence, creating and scheduling posts to increase engagement and visibility. - Recorded meeting minutes and coordinated chapter activities, supporting event planning and organizational operations.	
Lead instructor, Workshop on Geospatial Sciences <i>CoST Young Scientist Day - North Carolina A&T State University</i>	2025
- Led a Geospatial Sciences workshop, mentoring high school students on career pathways. - Demonstrated drones, GIS software, and 3D DTM technologies in hands-on sessions.	
Student Representative <i>BIOMES Seminar Committee - Yale University</i>	2022 – 2023
- Worked to invite speakers to speak at weekly Seminars and handle their logistics. - Helped to facilitate students' engagement with speakers.	
Co-Founder/Executive Director <i>Good Lead Foundation</i>	2019 – Present
- Leads the organization to initiate and implement projects. - Help raise funds to support educational activities within our community. - Organize seasonal internal capacity-building workshops for members of the Foundation.	

Certifications

- IBM Professional Data Engineering Certificate- Coursera (In Progress – Expected, September 2026)
- Foundations of AI Engineering – CodePath (2026)